Atharva Kale

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.97
Bharati Vidhyapeeth High School Haveli District Pune	Class XII (CBSE)	2021	96.00%
Bharati Vidhyapeeth High School Haveli District Pune	Class X (CBSE)	2019	88.60%

PROJECTS

Semantic-Based Attacks on 5G (MTECH THESIS) | OAI RAN, OAI Core

- Researched semantic-based attacks on 5G UEs, with existing approaches focusing on field-based constraints violation.
- Investigating capability-based constraint violations, with initial focus on 5G security capabilities as a novel approach
- Successfully demonstrated a attack by violating security capability on OAI 5G UE

Surface-Level Defect Detection of Assembly Line Products (BE Project) | Python, Flask, ReactJS

- Created an end-to-end system to detect/localize surface-level defects on assembly-line products, specifically metal sheets.
- Developed a dashboard using ReactJS and MySQL to store and visualize prediction results.
- Trained a YOLOv8 model on metal surface images and developed a Flask API to perform defect predictions.

Proxmox Private Server | Proxmox, Virtualization

- Built a homelab using Proxmox virtualization on repurposed hardware.
- Running various self-hosted services (NAS, BE project, personal projects) on bare-metal Proxmox having 99% uptime.
- Configured Tailscale as a private VPN for secure remote access to the home network from anywhere.

Restaurant Reviews Sentiment Analysis | Python, Flask, ReactJS

- Developed an NLP-based model to classify restaurant reviews as satisfactory or unsatisfactory.
- Built a Flask API to serve model predictions and integrated it with a ReactJS frontend.
- Deployed the application on a private server.

INTERNSHIPS

Cytel Software Services | Software Engineer Intern

- Collaborated with cross-functional teams of developers and QA testers to ship quality updates for a core legacy product.
- Developed a Proof of Concept (PoC) in C++ to extract and parse JSON parameters from CSV input, process the parameters, and invoke an R script.

ParamCS Software Services | Software Engineer Intern

- Developed an Employee Management System using C# .NET and SQL Server, supporting CRUD operations.
- Developed an Inventory Management System for production-based companies using C# .NET and SQL Server.

SKILLS

- **Languages:** C++, HTML/CSS, JavaScript, SQL
- **Frameworks:** ReactJS, LLVM

- **Developer Tools:** Docker, Kubernetes, Git, GitHub, Wireshark
- **Technologies:** Linux, Machine Learning

RELEVANT COURSES

Relevant Coursework: Advanced Computer Networks, Advanced Data Structures, Foundations of Machine Learning, Network Security, Wireless Networks, Introduction to Program Analysis and Compiler Optimization

POSITIONS OF RESPONSIBILITY

Secure Software Engineering | Teaching Assistant

- Designed and conducted a Web CTF focused on OWASP Top 10 web security vulnerabilities.
- Managed course assignments and evaluations with another TA.

Association of Computer Engineering Students | Technical Team Head

- Led the Technical Team of the Computer Engineering Student Committee and managed a team of 5 members.
- Organized and conducted various technical sessions for students.

Association of Computer Engineering Students | Web Team Joint Head

- Served as Joint Head of the Web Team of the Computer Engineering Student Committee and managed a team of 7.
- Migrated the committee website from HTML/CSS/JavaScript to ReactJS for improved performance and maintainability.

ACHIEVEMENTS

- Awarded **Best Outgoing Student** by the Computer Engineering Department for overall academic and extracurricular performance.

BLOGS

React Website Deployment on Virtual Machines

- Guide to deploying React applications on Linux virtual machines using Nginx and Proxmox (Medium).

Cloudflare Tunnels for Secured Public Access to Locally Deployed Websites

- Guide to securely exposing locally deployed websites to the public using Cloudflare Tunnels (Medium).

HOBBIES

Gaming, Cricket, Table Tennis, Mythology